

Serin Baek

CURRICULUM VITAE

London, United Kingdom | serin.baek2023@gmail.com | <https://www.linkedin.com/in/serin-baek-0a2bb2292/>

EDUCATION

University College London

MSci Pharmacology

2023 - present

First Class Honors (on track)

Awards: D.H. Jenkinson Prize, Dean's List

RESEARCH POSITIONS

Schiavo Group, Queen Square Institute of Neurology

September 2026

Supervised by Professor Giampietro Schiavo and Kai Sun

Investigated the structural interactions of cytoplasmic dynein-1 heavy chain with LIC1/LIC2 under conditions of serine phosphorylation and G-domain overexpression using Co-IP and Western Blot competitive binding assays.

Fitzgerald's Group, Early Cancer Institute, University of Cambridge

July 2026 - August 2026

Supervised by Professor Rebecca Fitzgerald and Dr. Ahsen Ustaoglu

Evaluated non-endoscopic capsule sponge assays and liquid biopsies for the early detection of oesophageal adenocarcinoma. Performed SIMOA and bioinformatic methylation analyses (EM-Seq vs. 5-Base) to identify and validate candidate protein (FABP2 and REG4) and epigenetic biomarkers.

Moss Group, UCL

June 2026 - September 2026

Supervised by Dr. Guy Moss and Dr. Thomas Blacker

Investigated gap-junctional coupling between airway epithelial cells and how CFTR-expressing cells may electrically support their non-expressing neighbours. Built a resistor-network computational model in R to fit published datasets and conducted FLIM using a genetically encoded voltage indicator (JEDI-2P) in HEK293T cells to measure membrane potential.

Sihra, UCL

September 2025 - June 2026

Supervised by Professor Talvinder Sihra

Analysed 82 peer reviewed studies, writing a 5000-word review on *Phosphorylation Dependent Modulation of Presynaptic Plasticity*, characterising a theoretical presynaptic multi-phosphocode, mapping how the temporal balance between competing kinase cascades encodes the direction and magnitude of synaptic plasticity.

Centre for Cardiovascular and Metabolic Neuroscience (CCMN), UCL

January 2025 - April 2025

Supervised by Professor Alexander Gourine and Dr. Alice Braga

Explored how astrocytic A2B receptor activation regulates brain energy metabolism and lactate release to support synaptic plasticity. Performed cryostat sectioning and immunostaining on murine brains to evaluate metabolic signalling pathways linked to cognitive health and neurodegeneration.

AWARDS AND FUNDING

Dean's List, UCL

August, 2026

Awarded in recognition of exceptional academic performance in the Faculty of Life Sciences (top 5%).

Early Cancer Institute Summer Research sponsored by Moderna

July, 2026

Fully funded an 8-week summer studentship at the Early Cancer Institute at the University of Cambridge.

Cystic Fibrosis Trust Summer Studentship Scheme, Cystic Fibrosis Trust UK

June, 2026

Fully funded a 7-week summer research on bioelectric profiling of airway epithelial cell types in cystic fibrosis.

D.H. Jenkinson Prize, UCL

September, 2024

Awarded to the top-performing student in the MSci/BSc Pharmacology cohort for best overall first-year academic performance (81% average).

Presidential Science Scholar (Top 20 Nationwide), South Korea

2023-2027

Awarded the most prestigious scholarship by the South Korean government, fully funding undergraduate studies abroad (approx. \$200,000 USD), with no bond requirement.

RESEARCH TALKS

Serin Baek, Thomas Blacker, and Guy Moss. Bioelectric Profiling and Ion Transport in Airway Epithelial Cells. (2026). Cystic Fibrosis Trust, September 23, London, United Kingdom.

Serin Baek, Ahsen Ustaoglu, Timothy Somerset, and Rebecca Fitzgerald. Translating Non-endoscopic Assays into Novel Diagnostics for Early Detection of Oesophageal Adenocarcinoma. (2026). Early Cancer Institute, August 27, Cambridge, United Kingdom.

WORK EXPERIENCE

Operational Excellence Summer Intern, AstraZeneca UK June 2025 - September 2025
Designed a Power BI dashboard for the UK-Sweden team to track KPIs, developed Excel VBA automation for SAP processes, and analysed 3,000+ Supplier Quality Risk Assessments to identify risk trends and support senior-level decision making. Received the You Made the Difference Award and a Gold certificate on Thriving in the age of AI.

LEADERSHIP EXPERIENCE

Student Ambassador, UK Council for International Student Affairs (UKCISA) December 2025 - present
Student ambassador of UK's national advisory board, providing lived-experience insight to improve support for international students in the UK. Invited to global education and policy forums as a guest speaker, including The PIE Live and UKCISA Annual Conference.

Room Monitor Supervisor, UCL October 2024 - present
Oversee all room monitors of UCL's administrative office to ensure operational readiness. Accumulated 600+ hours of early-morning room monitoring (8AM start), demonstrating consistency and diligence.

Student President, East-West International School, Cambodia August 2021 - Dec 2022
Initiated a volunteering partnership with Cambodia Children's Fund, designed 200+ student report card templates and developed Excel-based class timetables in collaboration with fellow interns.

VOLUNTEERING EXPERIENCE

Homeless Outreach, Euston Church October 2025 – May 2026
Baked desserts and spoke to homeless people every week, acting as the listening ears while gaining insight into how social vulnerability increases health challenges.

Volunteering and Welfare Officer, Music Society, UCL September 2024 - June 2025
Organised 4+ volunteering concerts at hospitals and schools, coordinating logistics and performance schedules with UCL Student Union and musicians.

Mobile Clinic Medical Volunteer, One Peninsula Medical Union, Cambodia July 2024
Supported clinicians in safe and effective dosing decisions, interpreting (Khmer) between doctors and patients to ensure clear communication.

TEACHING EXPERIENCE

East-West International School September 2022 – December 2022
Assistant teacher for A-Level Further Mathematics, as a valedictorian with an A* in A-Level Maths.

SKILLS

Programming: R, Python

Electrophysiology & Wet Lab: Western Blot, molecular cloning, dose-response assays, patch clamp, mammalian/neuronal cell culture, transfection, fluorescence microscopy, FLIM, SIMOA, ELISA, BCA Protein Assay

Imaging & Analysis: Andor Solis, Fiji (ImageJ), GraphPad Prism, HEKA Pulse, Excel